Paper Review: Smooth and Time Optimal Trajectory Planning for Industrial Manipulators along Specified Paths

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Motivation

- Industrial manipulators require high productivity while minimizing capital and operational costs.
- Consequently, the computation of time-optimal paths and trajectories is of significant interest.
- In many industrial applications, the geometric path is predefined; accordingly, this work considers the problem of path-constrained time-optimal trajectory planning.[1]
- The dynamics of robot manipulators are governed by highly nonlinear, coupled differential equations, rendering the problem challenging, particularly for systems with high degrees of freedom.

[1] A trajectory is defined as the time evolution of the robot's state.

Path-Constrained Time-Optimal Motion (PCTOM)

- Prior work has established that time-optimal trajectories exhibit bang-bang or bang-singular-bang control in actuator torques.
- Although time-optimal, such control laws are inherently discontinuous at switching points, leading to unbounded jerk.
- In practice, robot actuators cannot track abrupt changes in torque, resulting in tracking errors and lag relative to the reference trajectory.
- These tracking errors excite the feedback controller, often inducing vibrations and increased mechanical stress on the actuators.
- Persistent vibrations and stress can accelerate wear and potentially lead to premature failure of mechanical components.

- A typical PCTOM solution exhibits discontinuities and chattering behavior.
- Such effects can induce excessive wear and stress on actuators, limiting practical applicability.

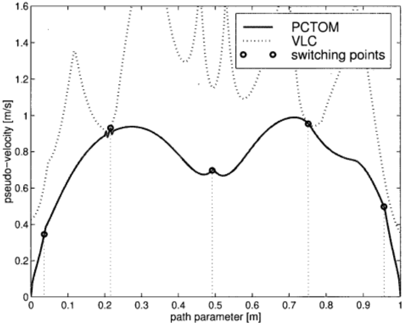


Figure: PCTOM in the s–$\dot{s}$ phase plane.

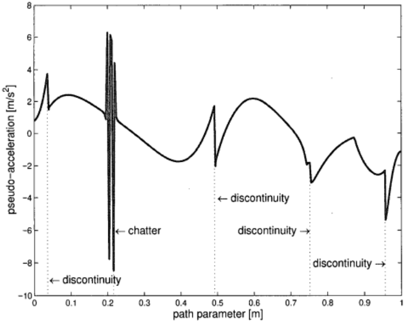


Figure: Chattering and discontinuities in PCTOM.

Smooth Path-Constrained Time-Optimal Motion (SPCTOM)

- This work addresses the limitations of PCTOM by incorporating actuator jerk constraints to obtain smooth, time-optimal trajectories.
- The inclusion of jerk constraints elevates jerk to the control input, resulting in a problem with both state and control constraints that may become active independently.
- Consequently, the optimal control structure no longer follows a simple bang-bang or bang-singular-bang form, and standard PMP-based TPBVP approaches are not directly applicable.
- The proposed approach reparameterizes the system using a scalar path variable and reformulates the time-optimal control problem as a nonlinear optimization problem in the phase plane.

Problem Formulation

Objective:[2]

$$\min_{\dot{T}\in\Omega} \int_0^{t_f} 1\, dt$$

subject to manipulator dynamics:

$$M(q)\ddot{q} + \dot{q}^T C(q)\dot{q} + G(q) = T$$

Boundary conditions:

$$q(0) = q_0, \quad q(t_f) = q_f$$

$$\dot{q}(0) = \dot{q}(t_f) = 0, \quad \ddot{q}(0) = \ddot{q}(t_f) = 0$$

Limits:

$$T_{\min} \leq T \leq T_{\max}, \quad \dot{T}_{\min} \leq \dot{T} \leq \dot{T}_{\max}$$

[2] PCTOM follows similar formulation without torque-rate constraints.

where

- $M(q) \in \mathbb{R}^{n\times n}$: inertia matrix
- $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$: Coriolis and centrifugal effects
- $G(q) \in \mathbb{R}^{n}$: gravity vector
- $q \in \mathbb{R}^{n}$: joint positions
- $T \in \mathbb{R}^{n}$: actuator torques
- $\dot{T} \in \mathbb{R}^{n}$: torque-rate (control input)

Since the control input is $\dot{T}$, the dynamics are differentiated to obtain:

$$M(q)\,\dddot{q} + \dot{M}(q)\ddot{q} + \ddot{q}^{T} C(q,\dot{q})\dot{q} + \dot{q}^{T}\dot{C}(q,\dot{q})\dot{q} + \dot{q}^{T} C(q,\dot{q})\ddot{q} + \dot{G}(q) = \dot{T}$$

Rewriting in state-variable representation, we get:

$$\dot{x} = F(x, u) = F(x) + K(x)u$$

$$x^T = (q^T, \dot{q}^T, \ddot{q}^T), \quad u = \dot{T}$$

where:

$$F(x) = \begin{bmatrix} \dot{q} \\ \ddot{q} \\ -M^{-1}(q)\Big(\dot{M}(q)\ddot{q} + \ddot{q}^T C(q)\dot{q} + \dot{q}^T \dot{C}(q)\dot{q} + \dot{q}^T C(q)\ddot{q} + \dot{G}(q)\Big) \end{bmatrix}$$

$$K(x) = \begin{bmatrix} 0 \\ 0 \\ M^{-1}(q) \end{bmatrix}$$

$$x_0 = (q_0, 0, 0), \quad x_f = (q_f, 0, 0)$$

To reduce the problem from 3n-dimesional space to 3, we parametrize the path via a scalar as:

$$r = r(s), \quad q = q(r(s))$$

Using the chain rule and simplifying:[3]

Torque constraint:

$$T = A(s)\ddot{s} + B(s)\dot{s}^2 + C(s)$$

$$T_{\min} \leq T \leq T_{\max}$$

where:

$$A(s) = M\,q'$$
$$B(s) = M\,q'' + q'^{T}\,C\,q'$$
$$C(s) = G$$

[3] $\dot{x} = \frac{dx}{dt}, \quad x' = \frac{dx}{ds}$

Torque-rate constraint:

$$\dot{T} = a(s)\,\dddot{s} + b(s)\dot{s}\ddot{s} + c(s)\dot{s}^3 + d(s)\dot{s}$$

$$\dot{T}_{\min} \leq \dot{T} \leq \dot{T}_{\max}$$

where:

$$\begin{aligned}
a(s) &= M\,q' \\
b(s) &= 3M\,q'' + \frac{dM}{ds}q' + 2\,q'^T\,C\,q' \\
c(s) &= M\,q''' + \frac{dM}{ds}q'' + q''^T\,C\,q' + q'^T\frac{dC}{ds}q' + q'^T\,C\,q'' \\
d(s) &= \frac{dG}{ds}\,\dot{s}
\end{aligned}$$

Torque Limits

- For each path position s, actuator torque limits define a feasible polygonal region in the $(\ddot{s}, \dot{s}^2)$ plane.
- These bounds impose constraints on pseudo-velocity and pseudo-acceleration:[4]

$$\dot{s} \leq \dot{s}^T_{\max}(s)$$

$$\ddot{s}^T_{\min}(s, \dot{s}) \leq \ddot{s} \leq \ddot{s}^T_{\max}(s, \dot{s})$$

- The curve $\dot{s}^T_{\max}(s)$ in the s–$\dot{s}$ plane is referred to as the **velocity limit curve (VLC)**.

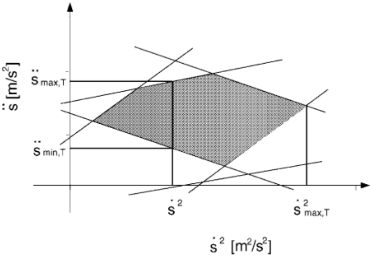


Figure: Admissible region in the $(\ddot{s}, \dot{s}^2)$ plane

[4] T denotes torque-induced bounds (as opposed to torque-rate constraints, denoted by J).

Torque-Rate Limits

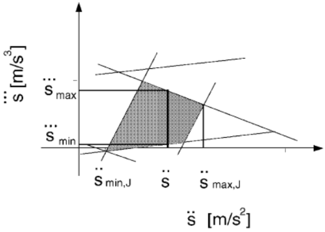
- For each $(s, \dot{s})$, torque-rate limits define a feasible region in the $(\dddot{s}, \ddot{s})$ plane.
- These impose bounds on pseudo-acceleration and pseudo-jerk:

$$\ddot{s}^J_{\min} \leq \ddot{s} \leq \ddot{s}^J_{\max}$$

$$\dddot{s}^J_{\min} \leq \dddot{s} \leq \dddot{s}^J_{\max}$$

- They also induce a velocity bound:

$$\dot{s} \leq \dot{s}^J_{\max}(s)$$



Admissible region in $(\dddot{s}, \ddot{s})$

- The limiting curve $\dot{s}^J_{\max}(s)$ occurs at points where:

$$\ddot{s}^J_{\min} = \ddot{s}^J_{\max}$$

- Combining torque and torque-rate constraints yields:

$$\min\left(\dot{s}^T_{\max}(s), \dot{s}^J_{\max}(s)\right)$$

- This defines the **smooth motion velocity limit curve (SMVLC)** in the s–$\dot{s}$ plane.
- The final bounds on pseudo-acceleration are obtained by taking the maximum of the lower bounds and the minimum of the upper bounds across all joints and both torque and torque-rate constraints. Similarly, global bounds on pseudo-jerk are obtained from torque-rate constraints. [5]

[5] Exact expressions are omitted for brevity; refer to the paper.

Reduced Scalar Dynamics

Following the reparameterization to scalar s, the control problem is formulated as:

State:

$$X = (s, \dot{s}, \ddot{s})^T, \quad u = \dddot{s}$$

Dynamics:

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} X + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$X_0 = (s_0, 0, 0), \quad X_f = (s_f, 0, 0)$$

Constraints:

$\dot{s}, \ddot{s}, \dddot{s}$ bounds from torque and torque-rate limits as discussed

Solution Approach

- Time-optimal control problems similar to SPCTOM are traditionally addressed using Pontryagin's Maximum Principle (PMP) with TPBVP formulations, or via dynamic programming and specialized algorithms.
- However, dynamic programming becomes computationally intractable due to exponential scaling with system dimension.
- Moreover, PMP-based approaches are not directly applicable, as the presence of both state and control constraints allows them to become active independently. Consequently, the optimal control does not exhibit a simple bang-bang or bang-singular-bang structure.

- To address this, the problem is reformulated in the s–$\dot{s}$ phase plane as a nonlinear parameter optimization problem.
- This reparameterization fixes the trajectory endpoints in the path domain, in contrast to the time domain where the final time remains free.
- The resulting objective is:

$$T_f = \int_{s_0}^{s_f} \frac{1}{\dot{s}(s)}\, ds$$

- **Optimization variables:**
    - $\dot{s}(s)$ along the path
    - Endpoint slopes $\frac{d\dot{s}}{ds}$

Trajectory Parameterization

- The trajectory in the s–$\dot{s}$ plane is inherently infinite-dimensional.
- We approximate it using **cubic splines** defined over fixed knot points.
- **Rationale:**
  - C^2 continuity ensures smooth acceleration profiles
  - Guarantees differentiability required for dynamics
  - Provides sufficient flexibility to approximate smooth trajectories
- Higher-order polynomials are avoided as they are unnecessary and may introduce oscillations.
- **Parameterization:**
  - $\dot{s}$ values at knot points
  - Boundary slopes $\frac{d\dot{s}}{ds}$ at endpoints
- Knot locations are fixed using switching points from the PCTOM solution, which serves as a good initialization since SPCTOM approaches PCTOM as torque-rate limits increase.

Optimization

- The optimization variables are normalized using the PCTOM solution:

$$\mathbf{x} = \left(\frac{\left.\frac{d\dot{s}}{ds}\right|_0}{\left.\frac{d\dot{s}}{ds}\right|_{m_0}}, \frac{\dot{s}_1}{\dot{s}_{m_1}}, \frac{\dot{s}_2}{\dot{s}_{m_2}}, \ldots, \frac{\dot{s}_p}{\dot{s}_{m_p}}, \frac{\left.\frac{d\dot{s}}{ds}\right|_f}{\left.\frac{d\dot{s}}{ds}\right|_{m_f}} \right)^T$$

- The problem is nonlinear and non-convex, with no closed-form expressions for the cost or constraints; therefore, a numerical evaluation-based approach is adopted.
- Since the optimal solution lies on the boundary of the feasible set, the optimization method must effectively handle both feasible and infeasible regions.
- To this end, a **Flexible Tolerance Method (FTM)** combined with **Nelder–Mead optimization** is employed[6]. In FTM, constraints are initially enforced with relaxed tolerances, which are progressively tightened as the optimization proceeds.

[6] Refer to the Nelder–Mead flexible polyhedron method for further details.

Typical Results

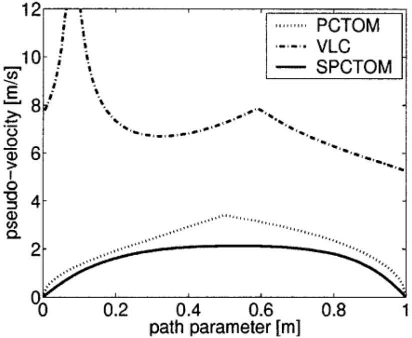


Figure: SPCTOM yields smoother $\dot{s}$ compared to PCTOM

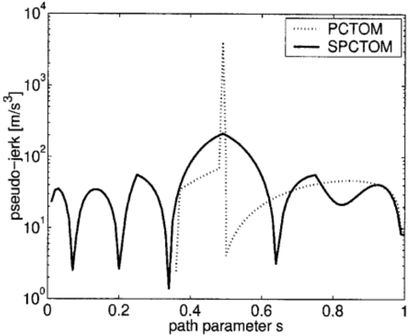
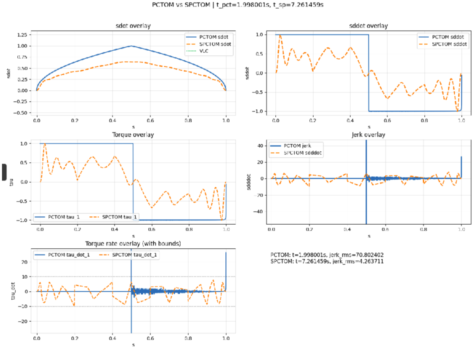


Figure: Torque-rate in SPCTOM is smooth and continuous

Results: Standard Car Problem

- The proposed framework is validated on a standard 1-DoF car problem.
- Although simple, this example provides a useful baseline for analysis.
- PCTOM exhibits impulsive jerk, whereas SPCTOM produces smooth trajectories that respect jerk bounds.



Results for 1-DoF car problem

Practical Challenges

- A feasible initialization is required. While trivial feasible solutions exist (e.g., very slow trajectories), identifying solutions close to the feasibility boundary is nontrivial.
- The FTM + Nelder–Mead approach relies on constraint violation measures. Its performance is sensitive to tuning parameters, which may lead to overly conservative solutions or imperfect constraint satisfaction.
- Convergence speed and solution quality depend strongly on initialization. In our implementation, the method often converges slowly and to local minima.
- The method depends on switching points obtained from PCTOM. We implemented the Shiller–Lu algorithm[7], which can be fragile, particularly near singular points, thereby affecting overall robustness.
- These observations suggest the need for more robust switching-point detection and improved optimization strategies that can reliably operate near the feasibility boundary.

[7] Refer to the original paper for details.

Evaluation

- The solutions are evaluated in terms of execution time, tracking accuracy, and smoothness.
- The comparative results are summarized in the table.
- Although jerk constraints may appear to increase traversal time, in practice, improved tracking reduces mismatch, resulting in competitive or lower execution times while mitigating wear and mechanical stress.

Trajectory	Torque rate limits [Nm/s]	Motion time [s]	Maximum tracking error [cm]	RMS tracking error joint 1 [°]	RMS tracking error Joint 2 [°]	RMS tracking error joint 3 [°]
PCTOM	∞	0.90	1.98	1.54	0.31	0.37
SPCTOM 2	100	0.74	1.40	1.12	0.26	0.33
SPCTOM 3	10	1.50	0.64	0.53	0.12	0.16

Simulated results for PCTOM and SPCTOM

Conclusion

- A methodology has been presented for computing smooth, time-optimal, path-constrained trajectories for robot manipulators.
- The proposed approach incorporates actuator jerk constraints to improve practical feasibility and reduce mechanical stress.
- The formulation relies on several assumptions and approximations; further work is required to explore more efficient optimization methods, robust switching-point detection, and improved strategies for operating near the feasibility boundary. [8]
- The code for the reimplemented framework can be found here: https://github.com/aadarshram/SPCTOM

[8] Some of the observations and limitations are based on our reimplementation.

References

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Thank You!